

RESEARCH REPORT

The Role of Attention for Context–Context Binding of Intrinsic and Extrinsic Features

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There is converging evidence that the feeling of conscious recollection is usually accompanied by the bound retrieval of context features of the encoding episode (e.g., Meiser, Sattler, & Weißer, 2008). Recently, however, important limiting conditions have been identified for the binding between context features in memory. For example, focusing on the semantics of the stimuli during encoding eliminates binding between perceptual context features (Meiser & Sattler, 2007). In the present research, we investigated the interplay of the focus of attention during encoding and stimulus characteristics in context–context binding. In particular, it has been suggested that context features differ in the degree to which they can be regarded as intrinsic or extrinsic to the items and that intrinsic features might be given more attentional processing during encoding than extrinsic features (e.g., Ecker, Zimmer, & Groh-Bordin, 2007a). In two experiments, we manipulated the “intrinsicity” of context features to investigate whether context–context binding might be limited to features that are in the focus of processing. Multinomial modeling analyses revealed that while context–context binding was eliminated for incidentally processed extrinsic context features (Experiment 1), it was preserved for intentionally processed extrinsic context features (Experiment 2).

Keywords: binding, source memory, remember/know, multinomial modeling

Episodic memories are inherently multidimensional, including a variety of different perceptual context features such as background color and shape or semantic features such as gender or academic status of a speaker. In addition, recent research has shown that episodic remembering goes together with a more integrated reinstatement of the encoding episode (Meiser & Bröder, 2002; Meiser, Sattler, & Weißer, 2008). This integrated retrieval of context details has been termed *context–context binding*, indicating that context features are bound to the item and to each other, forming a holistic representation of the feature configuration of the encoding episode (Meiser & Bröder, 2002).

In laboratory research, context–context binding is operationalized as stochastically dependent memory for context features.¹ Memory for context features is stochastically dependent if the probability of retrieving one context feature is higher when the other context feature is also retrieved than when the other context feature is not retrieved. For instance, Meiser and Bröder (2002) had participants study items that varied in position on the screen (upper vs. lower) and font size (large vs. small). Subsequently, participants’ memory was probed for their recognition of the items

and for the context features position and font size. In addition, participants were queried about whether they vividly recollected the prior presentation of the items (i.e., *remembered* [R]) or whether recognition was accompanied by a feeling of familiarity (i.e., *knew* [K]). Over a variety of studies, it has consistently been demonstrated that for items that are consciously recollected, retrieval of context features is stochastically dependent, whereas for items that are merely familiar, context features are retrieved independently (Meiser & Bröder, 2002; Meiser et al., 2008). This result is not due to higher overall levels of context memory in R responses than in K responses. Meiser et al. (2008) equated overall context memory between R and K responses experimentally and still observed context binding uniquely for R responses. This particular result also raises questions concerning the interpretation that R responses differ from K responses only in terms of overall memory strength. Although R/K judgments have also been explained by models assuming only one underlying process (see Wixted & Mickes, 2010, for a recent critique of the dual-process interpretation of RK data), the qualitative difference in context–context binding associated with R/K judgments cannot easily be accommodated by unidimensional models.

In line with the conceptualization of episodic memories as reinstatements of the encoding episode, the feeling of conscious recollection is thus associated with bound context memory of the encoding episode. Nonetheless, previous research has identified important lim-

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¹ Please note that the term *context* (or *context–context*) binding as used here refers to the phenomenon of stochastic dependency in memory for two context features, not to the binding of context features to the item.

iting conditions for context–context binding. For example, Meiser and Sattler (2007) showed that focusing on the meaning of stimulus words eliminated the binding between perceptual context features in R responses. In their study, one group of participants was asked to focus on the semantics of the words during study by embedding them in sentences, whereas another group was asked to focus on the word's perceptual features (i.e., screen location and font size). Results showed that binding between the perceptual features was characteristic for R responses under the perceptual encoding condition but not under the semantic encoding condition. Furthermore, Boywitt, Kuhlmann, and Meiser (in press) showed that aging forms a limiting factor for context–context binding. In this study, overall context memory was experimentally equated between young and older adults but context–context binding was characteristic of only young adults' R responses and was absent in older adults. Hence, there are important boundary conditions of context–context binding and investigating these can aid the understanding of the processes underlying episodic remembering. While previous research has focused on participant-inherent factors limiting context–context binding such as cognitive aging or processing style, we took a novel approach in our present research by looking at the interplay of attention and stimulus-inherent characteristics. Specifically, recent research suggests that context information differs in the degree to which it can be considered intrinsic or extrinsic to the item (e.g., Ecker et al., 2007a; Mulligan, 2011), and this factor may play a limiting role for the binding between context features (cf. Mather, 2007).

Intrinsic and Extrinsic Context Features

The features used so far to operationalize context are manifold and differ in the degree to which they can be considered part of item processing or more peripheral to the item. For example, investigators in some studies employed faces associated with target information to define context (e.g., Bayen & Murnane, 1996), whereas others used features such as position on the screen (e.g., Meiser & Bröder, 2002) or font size and color of the study stimuli (Boywitt et al., in press; Meiser et al., 2008; Mulligan, 2011), and yet others used semantic sources such as the gender (Meiser et al., 2008) or academic status of a speaker (Klauer & Wegener, 1998).

Recent research has drawn upon the idea that context features might generally differ in the degree to which they are item intrinsic or extrinsic (Ecker et al., 2007a; Ecker, Zimmer, & Groh-Bordin, 2007b). Ecker et al. (2007b), for example, defined intrinsic features as belonging to the study object and extrinsic features as features that are not part of the study object. A similar distinction was proposed by Spencer and Raz (1995) who distinguished between stimulus-bound (i.e., intrinsic) context and spatiotemporal (i.e., extrinsic) context. Ecker et al. (2007a) argued that intrinsic information influences familiarity, whereas extrinsic information drives recollective processes, presumably because intrinsic features can be processed automatically and extrinsic features require more attentional processing. Recently, Mulligan (2011) investigated the effect of generation instructions on memory for extrinsic and intrinsic context. In a series of experiments, participants were presented either with to-be-studied word pairs or with one word of the pair and had to generate the other word (i.e., the generation condition). Features such as font color or font type of the word pairs were used as intrinsic context and position of the study words on the screen served as extrinsic context. Results showed that memory for intrinsic context was hampered by generation instruc-

tions, whereas memory for extrinsic context was spared. Presumably, the generation instructions put semantic features of the stimuli into the focus of processing, and as a result, perceptual features are not given the same processing time or effort (Mulligan, 2004, 2011).

In line with this interpretation, we suggested that the focus of processing during encoding plays a critical role for context–context binding in memory. Specifically, features that could be considered rather intrinsic such as font color are often also closer to the focus of processing during study than features that could be considered rather extrinsic such as background color. For example, in order to encode a word, one must necessarily process the word in its specific color, whereas the background color must not necessarily be processed. Thus, even when not trying to encode the font color voluntarily, it is more likely to be processed than the background color. Accordingly, given that participants do not voluntarily try to encode context features along with the items, extrinsic context features receive less processing time and effort than intrinsic context features (cf. Mulligan, 2011; Spencer & Raz, 1995). On the other hand, if participants try to memorize both the study word and the color, it might not make a difference for context memory whether it is the color of the font or the color of the background that is to be memorized. In both cases, the color is put into the focus of processing by the explicit goal to memorize them together. Interestingly, Ecker et al. (2007a) found higher recognition performance for object–color pairings when color was intrinsic than when color was extrinsic even under intentional context encoding conditions. While this result indicates a recognition advantage for items with intrinsic features, the present study is rather concerned with context memory. To summarize, intrinsic features appear to be processed as a byproduct of item processing, whereas extrinsic features are not. Under conditions of intentional context learning, however, intrinsic and extrinsic features may be processed similarly.

There is evidence that context–context binding might rely on attentionally mediated processes and may thus be limited to context features processed in the immediate focus of attention such as intrinsic features. Uncapher, Otten, and Rugg (2006), for example, identified the intraparietal sulcus, an area thought to be involved in attentionally mediated binding of information, as specifically involved in encoding of multidimensional context features when these were later remembered jointly. Heightened activation in this area was uniquely associated with the joint retrieval of both context features, while the retrieval of the individual features of location and color was associated with other cortical regions. That is, successful bound encoding of context features appears to depend on a process additional to the encoding of the separate features (Uncapher et al., 2006). For the present research, this interpretation suggests that this binding mechanism might not be engaged for context features that are not in the immediate focus of attention such as extrinsic features.

Experiment 1

In order to investigate whether context binding is limited to intrinsic features, we assessed overall and bound context memory in an intrinsic and an extrinsic condition. Previous research investigating context–context binding employed features such as font size, position on the screen, font color, font type, and gender of the speaker (Meiser & Bröder, 2002; Meiser & Sattler, 2007; Meiser et al., 2008), features whose intrinsicity is debatable. For exam-

ple, some authors considered position as intrinsic (e.g., Mather, 2007), whereas others considered it as extrinsic (e.g., Mulligan, 2011). To avoid ambiguity regarding the intrinsicity of the context features and to avoid a confound between context condition and type of feature, we used the context features “color” and “position on the screen” in the intrinsic and the extrinsic condition, thus holding the type of feature constant across conditions.

Furthermore, study instructions did not mention that the context features would be relevant for the memory test, thus preventing participants from employing an explicit rehearsal strategy for the context features. To avoid floor effects in context memory we employed an orienting task that directed participants’ attention to the context features without asking them to memorize the features. This orienting task should lead participants to process the context features in a more peripheral way than the items while preserving context memory.

Method

Participants. One hundred and nine students participated for course credit or monetary compensation and were randomly assigned to one of the two context conditions. Data from nine participants were excluded from all analyses because they failed to complete the orienting task correctly.

Materials and design. From a set of 165 concrete German nouns, 64 words were randomly drawn as study items and 40 words were randomly drawn as distractors for the test phase for each participant anew.

The dimensions color (red vs. blue) and position on the screen (left vs. right) served as context features. In the intrinsic context condition, half of the 64 study items appeared to the left of a central black square and the other half appeared to the right of a central black square (dimension: position). Half of the items in each location appeared in red font, and the other half appeared in blue font (dimension: color). In the extrinsic context condition items were presented centrally on the screen in black font, with half of the items accompanied by a square to the left of the item and other half accompanied by a square to the right (dimension: position). Half of the squares on each side appeared in red, and the other half appeared in blue (dimension: color). Each combination of the dimensions position and color appeared equally often in an order that was randomized for each participant anew.

Procedure. Instructions, presentation of materials, and the memory test were computer directed. Without mentioning the relevance of the context, the initial instructions informed participants that they would be presented with a series of words that were to be memorized for a later memory test. For the orienting task, participants were instructed to check for every item on a response sheet in which position and color the word or the square was presented. Following the presentation of the instructions, participants were presented with the 64 study items one at a time. In the intrinsic condition, the presentation duration was 4 s with an interstimulus interval (ISI) of 1 s. In the extrinsic condition, the presentation duration was 5 s with a stimulus onset asynchrony of the colored square of 700 ms to ensure that participants were not distracted from the word.

After the study phase, participants completed an unrelated filler task for 3 min and then received instructions concerning the combined old/new, R/K, and context memory test. During the test phase, target items and distractors were displayed in random order

in the center of the screen in black font. For each item, participants indicated whether the word was old or new. If participants selected old, they indicated whether they remembered or knew the word. Then, participants indicated in which color and location the word or the square accompanying the word appeared during the study phase. All responses were made by selecting the respective response field on the screen.

Results

We set an alpha level of .05 for all analyses. Mean hit and false alarm rates of item recognition are displayed in Appendix A.

Overall context memory. In order to assess overall context memory, we computed the proportion of items correctly assigned to their respective context feature for each context dimension (i.e., color and position) separately. Table 1 displays these average conditional source-identification measure scores (ACSIM; Murnane & Bayen, 1996). Preliminary analyses showed that context memory for R responses was above chance level (i.e., higher than .5) in both conditions (see Table 1).

In the next step, we compared context memory for R and K responses across conditions. A mixed analysis of variance (ANOVA) with the repeated-measures factors judgment (R vs. K) and dimension (position vs. color) and the between-participants factor intrinsicity (intrinsic vs. extrinsic) revealed a main effect of judgment, $F(1, 93) = 27.15$, mean square error (MSE) = 0.026, $p < .001$, $\eta_p^2 = .23$, with better context memory for R than for K responses. Neither the main effect of dimension was significant, $F(1, 93) = 1.34$, $MSE = 0.016$, $p = .25$, $\eta_p^2 = .01$, nor the main effect of intrinsicity, $F(1, 93) < 1$. However, the interaction between judgment and intrinsicity was significant, $F(1, 93) = 6.33$, $MSE = 0.026$, $p = .014$, $\eta_p^2 = .06$, with a larger difference in context memory between R and K responses in the intrinsic condition than in the extrinsic condition. The interaction between judgment and dimension was not significant, $F(1, 93) = 3.42$, $MSE = 0.014$, $p = .068$, $\eta_p^2 = .04$.

Separate ANOVAs on context memory scores for each condition confirmed that context memory was significantly higher for R than for K responses in the intrinsic condition, $F(1, 45) = 24.43$, $MSE = 0.03$, $p < .001$, $\eta_p^2 = .35$, and in the extrinsic condition, $F(1, 45) = 4.53$, $MSE = 0.021$, $p = .038$, $\eta_p^2 = .09$.

Table 1

Mean Source Memory (Average Conditional Source Identification Measure) Scores (and Standard Errors) for Position and Color in Experiments 1 and 2

Condition	Experiment 1		Experiment 2	
	Intrinsic context	Extrinsic context	Intrinsic context	Extrinsic context
Remember				
Color	.59 (.02) ^a	.55 (.02) ^a	.71 (.02) ^a	.69 (.03) ^a
Position	.64 (.03) ^a	.57 (.02) ^a	.79 (.02) ^a	.70 (.03) ^a
Know				
Color	.48 (.02)	.53 (.02)	.57 (.02) ^a	.54 (.02) ^a
Position	.49 (.02)	.51 (.02)	.60 (.02) ^a	.58 (.02) ^a

^a Value significantly above chance level (i.e., .50), $p < .05$.

Model-based results. In order to assess stochastic dependency in context memory as an indication of context-context binding, we employed the reparameterized variant of the multinomial processing-tree model for two context dimensions (Meiser & Bröder, 2002). Please refer to the Appendix B for a figure. This model decomposes overall context memory into the contributions of independent and stochastically dependent (i.e., bound) context memory for R and K responses separately. Specifically, the model parameters d^R and d^K express the contribution of bound memory for both context dimensions over and above the contribution of independent memory for both context dimensions. Thus, the parameters d^R and d^K are direct measures of context–context binding. Independent context memory is captured by the parameters $e^{\text{Dim1 R}}$, $e^{\text{Dim1 K}}$, $e^{\text{Dim2 R}}$, and $e^{\text{Dim2 K}}$ for Context Dimensions 1 and 2, respectively, in case context features are not retrieved jointly (probability $1 - d^R$ and $1 - d^K$). The model parameters are process pure measures of the underlying cognitive processes and thus are preferred over behavioral measures that may be contaminated by nuisance processes such as guessing (Batchelder & Riefer, 1999).

Parameters estimation was completed on the basis of the aggregated observed response frequencies in each condition with the computer program multiTree (Moshagen, 2010), and overall model fit was assessed with the likelihood-ratio statistic G^2 (Batchelder & Riefer, 1999). The model provided a good fit to the data in the intrinsic condition, $G^2(18) = 12.62$, $p = .81$, and in the extrinsic condition, $G^2(18) = 24.22$, $p = .15$. Figure 1 displays the context–context binding parameter estimates. Please refer to the Appendix B for the other parameter estimates.

In the next step, we tested whether context–context binding was present in R and K responses in both conditions. The binding parameter for R responses (d^R) was significantly higher than the binding parameter for K responses (d^K) in the condition with intrinsic context, $G^2(1) = 8.67$, $p = .003$, but not in the condition with extrinsic context, $G^2(1) = 0.07$, $p = .80$. The crucial test, however, of whether context–context binding is present is whether the binding parameter is significantly different from zero. Indeed, the binding parameter was significantly larger than zero only for R responses in the intrinsic condition, $G^2(1) = 14.58$, $p < .001$, but not for K responses in the intrinsic condition, $G^2(1) = 0.0$, $p > .99$, and neither for R or K responses in the extrinsic condition, $G^2(1) = 0.98$, $p = .32$, and $G^2(1) = 0.21$, $p = .65$, respectively.²

Discussion

The results replicate the pattern of context–context binding for R but not for K responses in the intrinsic condition. In the extrinsic condition, however, context–context binding was completely eliminated. Overall context memory did not differ generally between both conditions and was above chance level for R responses in both conditions, ruling out that floor effects could account for the lack of context binding in the extrinsic condition. Although the difference in context memory between R and K responses was larger in the intrinsic condition than in the extrinsic condition, this difference does not explain the lack of context binding in the extrinsic condition as Meiser et al. (2008) ruled out that context–context binding might be a function of overall context memory.

One possible concern arising from the congruence of font color of the stimuli between the study phase and the test phase in the extrinsic condition (i.e., both black) is that studied items in this

condition might be more likely to be endorsed as old even if the context cannot be retrieved.³ Contrary to this concern, preliminary analyses revealed that overall recognition performance (in terms of hit rate – false alarm rate) did not differ between the intrinsic and the extrinsic condition, $t(98) = 1.02$, $p = .31$. According to this reasoning, context memory should be higher in the intrinsic than in the extrinsic condition. As shown previously, however, there are no overall differences in context memory between both conditions.

In sum, these results suggest that context–context binding is limited to intrinsic context features. It is important to note that this effect was not due to feature-specific differences between intrinsic and extrinsic features because the same features served as intrinsic context in one condition and extrinsic context in the other condition. Thus, features, which are not in the immediate focus of attentional processing during study, seem not to be bound in memory, presumably because an additional binding process was not engaged. This result is in line with the interpretation that context–context binding relies on an attentionally mediated binding mechanism that is engaged in addition to the processes involved in encoding the individual features (Uncapher et al., 2006). If this interpretation is correct, however, putting the extrinsic context features into the focus of attentional processing during study should produce context–context binding even for extrinsic features. This prediction was tested in the second experiment.

Experiment 2

To test whether putting the context features in the focus of attention during encoding produces context–context binding of extrinsic features, we employed the experimental setting from Experiment 1 but gave participants explicit instructions to encode the context along with the items. As opposed to the orienting task in Experiment 1, which led participants to process the context without the explicit goal to memorize it, the explicit context encoding instructions should ensure a level of attention to intrinsic and extrinsic context features sufficient for engaging the context binding mechanism.

Method

Participants. The 92 participants received either course credit or monetary compensation and were randomly assigned to one of the two conditions.

Materials and design. The materials and design were identical to those in the first experiment.

Procedure. The procedure was identical to the first experiment with the following exception. The orienting task was replaced by the explicit instructions to memorize the position and color of the words or the position and color of the simultaneously presented square along with the words. The presentation duration in both conditions was 4 s with an ISI of 1 s.

² If a parameter is set equal to the boundary of the parameter space (i.e., 0 or 1), the assumptions leading to an asymptotic chi-square distribution of the likelihood-ratio statistic G^2 cannot strictly be assumed. Alternatively, one can interpret the confidence intervals which in all cases but in R responses in the intrinsic context condition overlap with zero and thus yield the same conclusions.

³ We thank Ullrich Ecker for pointing out this possibility.

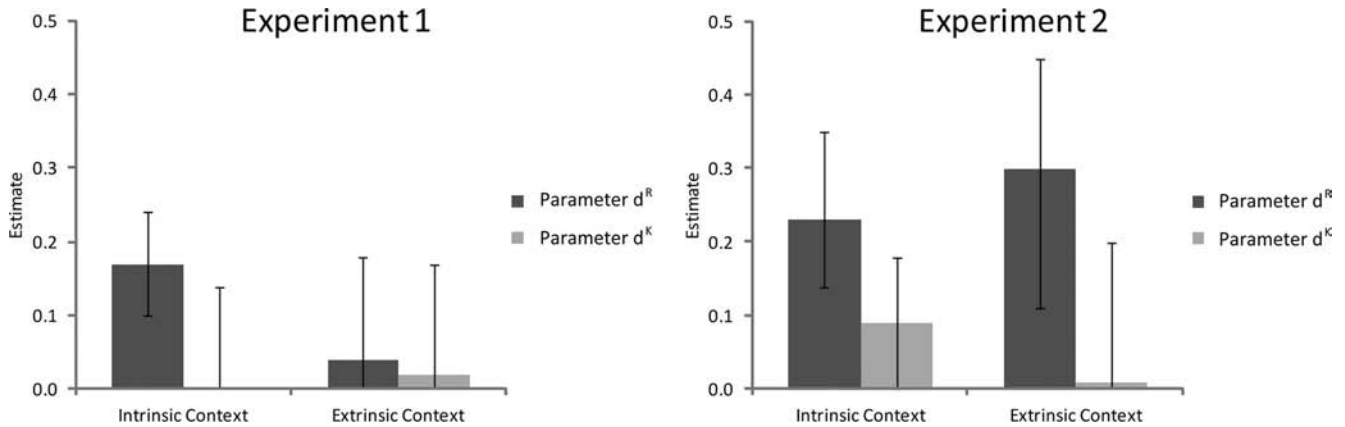


Figure 1. Binding parameters for remember (d^R) and know responses (d^K) in Experiments 1 and 2 estimated with the reparameterized variant of the multinomial model of multidimensional source memory. Error bars represent 95% confidence intervals.

Results and Discussion

Data from one participant were excluded from all further analyses because of negative item recognition performance in terms of hit rate minus false alarm rate. Mean hit and false alarm rates of item recognition are displayed in Appendix A.

Overall context memory. Overall context memory scores are presented in Table 1. A repeated-measures ANOVA on the ACSIM scores revealed a main effect of judgment, $F(1, 82) = 72.56$, $MSE = 0.025$, $p < .001$, $\eta_p^2 = .47$, with higher context memory scores for R than for K responses. The main effect of context dimension was also significant, $F(1, 82) = 6.02$, $MSE = 0.021$, $p = .016$, $\eta_p^2 = .068$, with better context memory for position than for color. Furthermore, there was a nonsignificant tendency of better context memory for intrinsic than for extrinsic context features, $F(1, 82) = 3.41$, $MSE = 0.04$, $p = .068$, $\eta_p^2 = .04$. None of the interactions was significant, all $ps > .16$. Thus, memory for context was equally higher for R than for K responses in both conditions.

Model-based results. The model provided a good fit to the data in the intrinsic condition, $G^2(18) = 6.40$, $p = .99$, and in the extrinsic condition, $G^2(18) = 19.36$, $p = .37$. Figure 1 displays the context–context binding parameter estimates.

Analysis of context–context binding revealed that the binding parameter for R responses was significantly higher than the binding parameter for K responses in the extrinsic condition, $G^2(1) = 9.79$, $p = .002$, and marginally higher in the intrinsic condition, $G^2(1) = 1.98$, $p = .16$. The more powerful test of the binding parameters against zero, however, yielded significance for R responses in the extrinsic condition, $G^2(1) = 22.35$, $p < .001$, and in the intrinsic condition, $G^2(1) = 8.00$, $p = .005$, but not for K responses in either condition: extrinsic, $G^2(1) = 0.02$, $p = .89$, and intrinsic, $G^2(1) = 1.11$, $p = .29$. Furthermore, the binding parameters for R or for K responses did not differ between conditions, $G^2(1) = 0.97$, $p = .32$, and $G^2(1) = 0.40$, $p = .53$, respectively.

These results confirm that in the case of intentional context encoding, context–context binding was similarly present in R responses in the condition with intrinsic and in the condition with

extrinsic context features but was absent in K responses in both conditions. This finding is in line with the prediction that changing the focus of processing during encoding puts the extrinsic context features into the attentional focus, and these features are then also retrieved jointly in the case of R responses.

General Discussion

Episodic remembering is usually characterized by the integrated retrieval of contextual details from the encoding episode (e.g., Meiser et al., 2008), but the present study suggests that as a necessary precondition, the context features must be in the processing focus at time of encoding. Context features that are more peripheral to the focus of processing were not retrieved in a bound fashion. In line with earlier accounts (e.g., Ecker et al., 2007a; Spencer & Raz, 1995), we suggest that extrinsic context features are not spontaneously given the same processing time or effort as intrinsic context features and consequently do not benefit from attentionally mediated binding processes (cf. Uncapher et al., 2006). Corroborating this interpretation, our first experiment showed that despite similar levels of overall context memory, context–context binding was eliminated in the extrinsic condition but preserved in the intrinsic condition. Those features that are closer to the focus of attention during encoding, such as intrinsic features, can be stored in a bound fashion, whereas those features that are more peripheral to the focus of attention, such as extrinsic features, are not stored in a bound fashion. Altering the focus of processing, however, should result in bound storage of context features regardless of whether they are intrinsic or extrinsic to the item. In line with this prediction, explicit instructions to encode the context features along with the item in the second experiment produced binding even among extrinsic context features. This result is also in line with the finding by Mulligan (2011) that putting extrinsic features into the focus of processing alters the effect of generation instructions on context memory. Thus, altering the focus of attention appears to alter the encoding strategy so that intrinsic and extrinsic features are processed similarly to the item, that is, in an explicit fashion. For example, participants might adopt an explicit encoding strategy such that they recode the features position and color into verbal labels (e.g., “left,” “green”), alleviating the intrinsic/extrinsic

difference.⁴ In sum, the present results corroborate the notion that extrinsic context is not spontaneously attended to during encoding and therefore is not bound in memory but altering the focus of attention leads to bound context memory even for extrinsic features, although this does not happen spontaneously.

This finding adds to the accumulating evidence pointing to the involvement of an attentionally mediated encoding mechanism combining the context features into a coherent memory representation (Uncapher et al., 2006). This process could be conceptualized as a binding process operating in the closer focus of attentional processing, which integrates the information into a coherent memory trace. Then, the later retrieval of the memory trace leads to the bound retrieval of context features and the concomitant subjective experience of conscious recollection. In line with introspection, this interpretation suggests that episodes that especially attract our attention are later remembered very vividly, likely because the heightened attention during encoding allowed for the integrative storage and retrieval of the context features. Notably, the present view shifts the focus from the processes operating at retrieval to the processes operating at encoding for the subjective experience of remembering.

More generally, this result also has important implications for the conceptualization of context. For instance, giving explicit source encoding instructions as opposed to incidental encoding instructions in studies of source memory might alter the processes one aims to study. For example, Boywitt et al. (in press) did not observe context–context binding in older adults with incidental source encoding instructions. Possibly, older adults have a more limited attentional focus than young adults (Craik & Byrd, 1982) and features that are relatively close to the focus of processing for young adults are more peripheral to the focus of processing for older adults and thus are not bound in memory. Giving explicit source encoding instructions might put those features into the focus of processing even for older adults and consequently might result in context–context binding. In this case, however, explicit source encoding instructions would change the processes that participants engage in, and researchers should be aware of this when giving explicit source encoding instructions.

It is further noteworthy that the conclusions drawn here rest on the results obtained with visuospatial features. Previous research, on the other hand, has demonstrated context–context binding of semantic features such as the gender of a speaker (Meiser et al., 2008), and the intrinsicity of these feature is not immediately clear. For example, while combinations of two extrinsic features did not automatically produce context–context binding and combinations of two intrinsic features resulted in binding, combinations of one extrinsic feature and one intrinsic feature might take an intermediate position (see also Mather, 2007).

Overall, in the present research, we took a novel approach by extending the knowledge of boundary conditions for the joint retrieval of context features from episodic memory to the role of attentional focus during encoding, and thus our findings provide further insights into the processes involved in episodic remembering.

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⁴ We thank Ullrich Ecker for this suggestion.

Appendix A

Table A1

Mean Proportions (and Standard Errors) of Remember and Know Responses to Target Items (Hits) and Distractors (False Alarms) in Experiments 1 and 2

	Experiment 1		Experiment 2	
	Intrinsic context	Extrinsic context	Intrinsic context	Extrinsic context
Hits				
Remember	.33 (.03)	.37 (.02)	.39 (.03)	.39 (.03)
Know	.26 (.02)	.24 (.02)	.32 (.02)	.24 (.02)
False alarms				
Remember	.03 (.01)	.03 (.01)	.01 (.003)	.03 (.01)
Know	.13 (.02)	.12 (.02)	.11 (.02)	.10 (.01)

(Appendices continue)

Appendix B

Multinomial Model of Multidimensional Source Memory

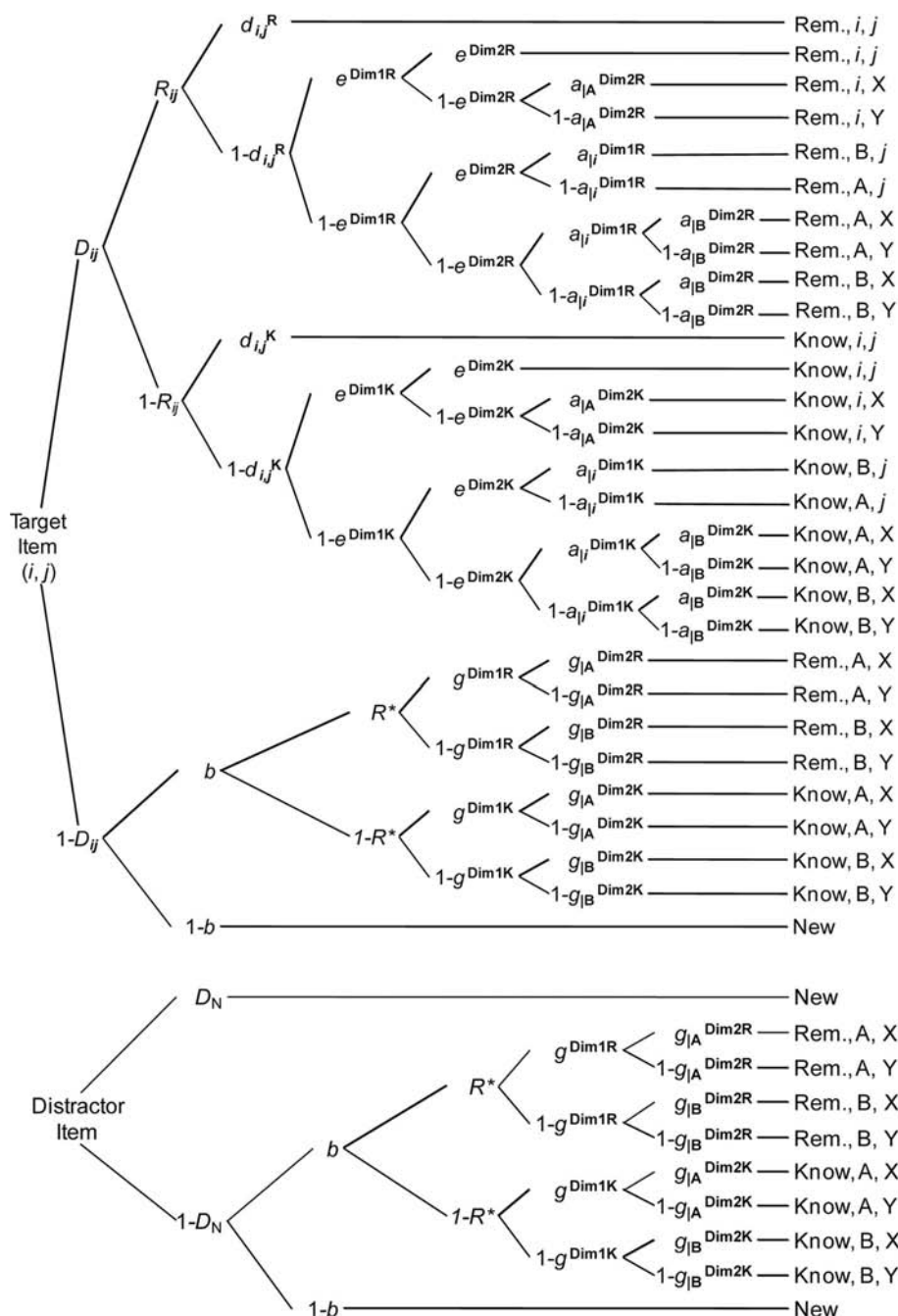


Figure B1. Processing-tree diagram of the reparameterized variant of the multinomial model of multidimensional source memory (Meiser & Bröder, 2002). Adapted from "The Role of Source Memory in Older Adults' Recollective Experience" by C. D. Boywitt, B. G. Kuhlmann, and T. Meiser (in press), *Psychology and Aging*. Adapted with permission. Copyright 2011 by the American Psychological Association.

(Appendices continue)

Table B1

Parameter Estimates [and 95% Confidence Intervals] Derived With the Reparameterized Variant of the Multinomial Model of Multidimensional Source Memory

Parameter	Experiment 1		Experiment 2	
	Intrinsic context	Extrinsic context	Intrinsic context	Extrinsic context
D	.43 [.41, .46]	.47 [.45, .49]	.60 [.57, .62]	.51 [.49, .53]
R	.70 [.67, .75]	.73 [.70, .77]	.71 [.67, .74]	.64 [.61, .67]
R^*	.17 [.13, .21]	.21 [.16, .25]	.08 [.05, .12]	.22 [.16, .28]
b	.28 [.25, .30]	.28 [.26, .30]	.29 [.26, .31]	.23 [.21, .26]
d^R	.17 [.09, .24]	.04 [.00, .11] ^a	.23 [.10, .35]	.30 [.21, .39]
d^K	.00 [.00, .14] ^a	.02 [.00, .17] ^a	.09 [.00, .24] ^a	.01 [.00, .20] ^a
$e^{\text{position R}}$	.15 [.05, .25]	.10 [.01, .19]	.48 [.37, .58]	.21 [.09, .33]
$e^{\text{position K}}$	.01 [.00, .20] ^a	.00 [.00, .21] ^a	.29 [.13, .44]	.24 [.04, .43]
$e^{\text{color R}}$	.00 [.00, .11] ^a	.07 [.00, .17] ^a	.30 [.17, .43]	.25 [.14, .37]
$e^{\text{color K}}$	.00 [.00, .19] ^a	.11 [.00, .30] ^a	.14 [.00, .32] ^a	.19 [.00, .39] ^a

Note. D = probability of recognizing a study word; R = probability of assigning a remember (R) response to a recognized studied word; R^* = probability of assigning a R response to distractor words that were guessed to be “old”; b = probability of classifying an undetected distractor as old; d^R = probability of bound context memory for study words classified as R; d^K = probability of bound context memory for study words classified as know (K); e = independent memory for the context features. Superscript indicates context dimension and classification of memory experience.

^a Confidence intervals with limits of .00 were bounded by the parameter space that limits the range of each model parameter to the interval [.00, 1.00].

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